

Jianfeng Ning

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Short Bio

Jianfeng Ning (male, b. 1997) is a Postdoctoral researcher in the group Nonsmooth Variational Problems and Operator Equations at Weierstrass Institute in Berlin. His research interests include numerical PDEs, inverse problems, optimization and machine learning.

Education and Experiences

Weierstrass Institute for Applied Analysis and Stochastics

Postdoctoral Researcher

Germany

Since 2025.09

Wuhan University

Ph.D in Computational Mathematics

China

2019.09–2025.06

○ 2023.10–2025.06: Research Assistant, The Chinese University of Hong Kong, Hong Kong

○ 2022.02–2023.09: Visiting Student, University of Bremen, Germany

Lanzhou University

B.Sc in Mathematics and Applied Mathematics

China

2015.09–2019.06

Publications

2026: Michael Hintermüller and **Jianfeng Ning**. “Constrained Neural Parameterization for Optimization in Function Spaces.” arXiv preprint arXiv:2606.00855, 2026.

2025: **Jianfeng Ning**, and Jun Zou. “Constructing probing functions for direct sampling methods for inverse scattering problems with limited-aperture data.” Journal of Computational Physics 528(2025) 113820.

2025: **Jianfeng Ning**, Fuqun Han and Jun Zou. “A Direct Sampling Method and Its Integration with Deep Learning for Inverse Scattering Problems with Phaseless Data.” SIAM Journal on Scientific Computing 47(2025) pp. C343-C368.

2025: Derick Nganyu Tanyu, **Jianfeng Ning**, Andreas Hauptmann, Bangti Jin and Peter Maass. “Electrical Impedance Tomography: A Fair Comparative Study on Deep Learning and Analytic-based Approaches.” Data-driven Models in Inverse Problems, De Gruyter, 2025, pp. 437-470.

2025: Alexander Denker, Fabio Margotti, **Jianfeng Ning**, Kim Knudsen, Derick Nganyu Tanyu, Bangti Jin, Andreas Hauptmann, Peter Maass. “Deep learning based reconstruction methods for electrical impedance tomography.” Handbook of Numerical Analysis. Amsterdam: Elsevier, 2025, 26: 127-175.

2024: **Jianfeng Ning**, Fuqun Han and Jun Zou. “A direct sampling-based deep learning approach for inverse medium scattering problems.” Inverse Problems 40(2024) 015005.

2023: Derick Nganyu Tanyu, **Jianfeng Ning (Co-first author)**, Tom Freudenberg, Nick Heilenkötter, Andreas Rademacher, Uwe Iben and Peter Maass. “Deep learning methods for partial differential equations and related parameter identification problems.” Inverse Problems 39(2023) 103001.

Professional Activities

Member: Inverse Problems International Association (IPIA)

Reviewer: SIAM Journal on Numerical Analysis, SIAM Journal on Scientific Computing, Inverse Problems, Machine Learning: Science and Technology, Journal of Franklin Institute, Physica Scripta

Honors & Awards

2026: EAIP PhD Thesis Award

2025: Outstanding Graduate Award, School of Mathematics and Statistics, Wuhan University

2025: First-Class Academic Innovation Scholarship for Postgraduate Students, Wuhan University

2024: Second-Class Academic Innovation Scholarship for Postgraduate Students, Wuhan University

2020: Excellent Teaching Assistant, Wuhan University

2020: Second-Class Scholarship, Wuhan University

2019: First-Class Scholarship for Graduate Freshman, Wuhan University

2019: Outstanding Graduate Award, Lanzhou University

2017–2018: First-Class Scholarship, Lanzhou University

2016: China National Scholarship

Skills

Programming skills: Python, PyTorch, MATLAB, C++

Knowledge of languages: Chinese (first language), English (professional working proficiency)